

Chapter 7. CMake build functions

7.1. Control of picotool post-processing (not available on RP2040)

These functions control what post-processing is performed by picotool on the executables produced by the sdk. They can be used to hash/sign binaries, package no_flash binaries, and embed partition tables into block loops. These functions all set the specified target properties, and therefore if the property should be set for multiple targets then it can be set manually with `set_property` or other CMake functions to set properties for a given scope.

<code>pico_package_uf2_output(TARGET PACKADDR)</code>	Package a UF2 output to be written to the PACKADDR address. This can be used with a no_flash binary to write the UF2 to flash when dragging & dropping, and it will be copied to SRAM by the bootrom before execution. This sets <code>PICOTOOL_UF2_PACKAGE_ADDR</code> to PACKADDR.
<code>pico_set_otp_key_output_file(TARGET OTPFILE)</code>	Output the public key hash and other necessary rows to an otp JSON file. This sets <code>PICOTOOL_OTP_FILE</code> to OTPFILE.
<code>pico_load_map_clear_sram(TARGET)</code>	Adds an entry to the load map to instruct the bootrom to clear all of SRAM before loading the binary. This appends the <code>--clear</code> argument to <code>PICOTOOL_EXTRA_PROCESS_ARGS</code> .
<code>pico_set_binary_version(<TARGET> [MAJOR <version>] [MINOR <version>] [ROLLBACK <version>] [ROLLBACK_ROWS <rows...>])</code>	Adds a version item to the metadata block, with the given major, minor and rollback version, along with the rollback rows. These are appended as arguments to <code>PICOTOOL_EXTRA_PROCESS_ARGS</code> if setting the rollback version, or set as compile definitions if only setting the major/minor versions.
<code>pico_set_uf2_family(TARGET FAMILY)</code>	Set the UF2 family to use when creating the UF2. This sets <code>PICOTOOL_UF2_FAMILY</code> to FAMILY.
<code>pico_sign_binary(TARGET [SIGFILE])</code>	Sign the target binary with the given PEM signature. This sets <code>PICOTOOL_SIGN_OUTPUT</code> to true, <code>PICOTOOL_SIGFILE</code> to SIGFILE (if specified), and <code>PICOTOOL_OTP_FILE</code> to <code>\${TARGET}.otp.json</code> (if not already set). To specify a common SIGFILE for multiple targets, the SIGFILE property can be set for a given scope, and then the SIGFILE argument is optional.
<code>pico_hash_binary(TARGET)</code>	Hash the target binary. This sets <code>PICOTOOL_HASH_OUTPUT</code> to true.
<code>pico_embed_pt_in_binary(TARGET PTFILE)</code>	Create the specified partition table from JSON, and embed it in the block loop. This sets <code>PICOTOOL_EMBED_PT</code> to PTFILE.
<code>pico_encrypt_binary(TARGET AESFILE [SIGFILE])</code>	Encrypt the target binary with the given AES key (should be a binary file containing 32 bytes of a random key), and sign the encrypted binary. This sets <code>PICOTOOL_AESFILE</code> to AESFILE, and <code>PICOTOOL_ENC_SIGFILE</code> to SIGFILE if present, else <code>PICOTOOL_SIGFILE</code> .